Arash Jalil Khabbazi, MS

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Profile

I work on machine learning, optimization, and advanced control algorithms to make buildings and energy systems smarter and more efficient. I apply experience from both research and industry to solve practical engineering problems.

Education

Purdue University, *IN, United States* — 2023 –
PhD in Mechanical Engineering — **GPA:** 3.84/4.0
Minor in Computational Science and Engineering
Advisor: Kevin J. Kircher

University of British Columbia, *BC, Canada* — 2021 – 2023
MS in Mechanical Engineering — **GPA:** 4.33/4.33 (94%)
Thesis: *Mixing gaseous hydrogen into natural gas distribution pipelines*
Advisor: Sunny R. Li

University of Tabriz, *EA, Iran* — 2016 – 2020
BS in Mechanical Engineering — **GPA:** 4.0/4.0 (19.12/20) — **Rank:** 1/155+

Publications — Full list on Google Scholar

Journal articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, "Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review," *Applied Energy*, 2025. [DOI]

1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, "Mixing hydrogen into natural gas distribution pipeline system through Tee junctions," *International Journal of Hydrogen Energy*, 2024. [DOI]

Conference papers

6. **A.J. Khabbazi** and K.J. Kircher, "Do Small HVAC Control Demonstrations in Larger Buildings Risk Overestimating Savings?," *under review*, 2025.

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5. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, "Active current limiting control of residential appliances for breaker panel protection across the US: a parametric study," *CISBAT*, 2025, pp. 1–6.

4. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, K.J. Kircher, "What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?," *International High Performance Buildings Conference*, 2024, pp. 1–10. [DOI]

3. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, "Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions," *Canadian Society for Mechanical Engineering International Congress*, 2023, pp. 1–6. [DOI]

2. **A. Khabbazi**, R. Li, and J. Quinn, "Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid," *Canadian Society for Mechanical Engineering International Congress*, 2022, pp. 1. [DOI]

1. **A. Khabbazi**, R. Li, and J. Quinn, "The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines," *International Green Energy Conference (IGEC-XIII)*, 2021, pp. 1. [DOI]

Industry experience

Summer Intern, Tesla, *Reno, NV, United States* — 05/2025 – 08/2025

- Built supervised ML models (XGBoost, neural nets, SVR; ensembles) for defect detection and quality assessment.
- Supported deployment of a thermal imaging system for non-destructive inspection in drive-unit manufacturing.
- Built data pipelines with MySQL and applied feature engineering to identify high-signal inputs for ML development.

- Contributed to projects projected to reduce testing costs by over $1M annually through improved quality assessment.

Student Researcher, FortisBC, *Vancouver, BC, Canada* 09/2021 – 09/2023

- Supported assessments on hydrogen injection into natural gas pipelines and renewable gas projects (UBC H_2Lab).
- Collaborated with engineers and research teams (materials, sensors, AI, safety) to assess industry impacts.

Research experience

PhD Research Assistant, Purdue University 09/2023 –

- Supported commissioning of Herrick Labs by upgrading the BAS (Tridium Niagara) and integrating Modbus systems.
- Developed ML models for occupancy detection (98% accuracy) and real-time energy/comfort dashboards (InfluxDB).
- Designed HVAC control strategies with MPC and DeePC, showing DeePC matched MPC without a system model.
- Reviewed 100+ publications on MPC and RL in HVAC, leading to a published review paper in Applied Energy.

MS Research Assistant, University of British Columbia 07/2021 – 09/2023

- Researched hydrogen injection into natural gas pipelines using CFD, real-gas modeling (C), and CAD design.
- Analyzed 1 TB of simulation data, leading to a journal paper, three conference papers, and two awards.

Skills

Programming: Python, C, MATLAB, R, SQL (MySQL), Git, Linux
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Keras
Data & Analytics: NumPy, Pandas, SciPy, Matplotlib, Seaborn, Grafana
Optimization & Modeling: CVX, CVXPY
Systems: Tridium Niagara (BAS), InfluxDB (TSDB), Modbus, IO-Link, MTConnect

Selected Honors & Awards

- ASHRAE Graduate Student Grant-in-Aid Award 2025
- Best Paper Award at CSME 2023 Conference 2023
- Best Presentation Award at CSME 2022 Conference, Advanced Energy Symposium 2022
- UBC Graduate Scholarship 2022
- UBC Dean's Entrance Scholarship 2021

Conferences

8th International High Performance Buildings Conference, *West Lafayette, USA* 07/2024
ASHRAE Winter Conference, *Chicago, USA* 01/2024
Canadian Society for Mechanical Engineering 2023 Conference, *Sherbrooke, Canada* 05/2023
Canadian Society for Mechanical Engineering 2022 Conference, *Edmonton, Canada* 07/2022
13th International Green Energy Conference, *Virtual* 07/2021

Workshops & seminars

Intelligent Building Operations (IBO) Workshop, *West Lafayette, USA* 07/2024
What have we learned from field demonstrations of advanced commercial HVAC control? [Recording]

Selected Courses

Applied Mathematics & Data Science: Artificial Intelligence, Applied Machine Learning, Statistical Methods, Numerical Computations

Optimization, Control & IoT: Reinforcement Learning and Control, Reinforcement Learning, Convex Optimization, Industrial IoT Implementation

Thermal & Energy Systems: Distributed Energy Resources, Analysis of Thermal Systems, Advanced Thermodynamics, Refrigeration Systems, Power Plants